

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

1) Create dept table and insert records

```
insert into dept_rollno values(10,'Account','Bangalore',702);
```

```
insert into dept_rollno values(20,'Sales','Delhi',705);
```

```
insert into dept_rollno values(30,'Research','Pune',707);
```

```
insert into dept_rollno values(40,'Developing','Hyderbad',null);
```

```
SQL> create table dept_22co11
  2  (Did number(3) primary key, Dname varchar(20),Loc varchar(15),Mgrid number(3));
```

Table created.

```
SQL> describe dept_22co11;
```

Name	Null?	Type
-----	-----	-----
DID	NOT NULL	NUMBER(3)
DNAME		VARCHAR2(20)
LOC		VARCHAR2(15)
MGRID		NUMBER(3)

```
SQL> insert into Dept_22co11 values(10,'Accounts','Bangalore',702);
```

1 row created.

```
SQL> insert into Dept_22co11 values(20,'Sales','Delhi',705);
```

1 row created.

```
SQL> insert into Dept_22co11 values(30,'Research','Pune',707);
```

1 row created.

```
SQL> insert into Dept_22co11 values(40,'Developing','Hyderbad',null);
```

1 row created.

```
SQL> select * from dept_22co11;
```

DID	DNAME	LOC	MGRID
-----	-----	-----	-----
10	Accounts	Bangalore	702
20	Sales	Delhi	705
30	Research	Pune	707
40	Developing	Hyderbad	

2) Create emp table

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
SQL> create table Emp_22co11
  2  (EID number(3) primary key, Name varchar(15), Salary number(6), Hire_date date, Job varchar(15), Did number(3) references dept_22co11(did), Mid number(3));
Table created.
```

```
SQL> describe emp_22co11;
```

Name	Null?	Type
EID	NOT NULL	NUMBER(3)
NAME		VARCHAR2(15)
SALARY		NUMBER(6)
HIRE_DATE		DATE
JOB		VARCHAR2(15)
DID		NUMBER(3)
MID		NUMBER(3)

3) Alter the emp table by making mid foreign key referring to eid primary key of emp table

```
SQL> alter table emp_22co11
  2  add constraint fk_DID foreign key(MID) references emp_22co11(EID);
Table altered.
```

4) Insert the records in emp table

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
SQL> insert into Emp_22co11 values(707,'Vishal',9500,'01-Jan-2001','Manager',30,null);
1 row created.

SQL> insert into Emp_22co11 values(705,'Lalit',6500,'08-Oct-2002','Manager',20,707);
1 row created.

SQL> insert into Emp_22co11 values(701,'Deepak',8000,'05-Jan-2001','Analyst',30,707);
1 row created.

SQL> insert into Emp_22co11 values(702,'Naresh',9000,'10-Jan-2001','Manager',10,707);
1 row created.

SQL> insert into Emp_22co11 values(703,'Sumesh',7000,'05-Feb-2001','Salesman',20,705);
1 row created.

SQL> insert into Emp_22co11 values(704,'Aditya',9000,'27-Nov-2007','Analyst',30,707);
1 row created.

SQL> insert into Emp_22co11 values(706,'Amit',null,'04-Nov-2004','Clerk',10,702);
1 row created.

SQL> insert into Emp_22co11 values(708,'Sumit',8000,'06-Jan-2006','Accountant',10,702);
1 row created.
```

```
SQL> set linesize 160;
SQL> select * from emp_22co11;
```

EID	NAME	SALARY	HIRE_DATE	JOB	DID	MID
707	Vishal	9500	01-JAN-01	Manager	30	
705	Lalit	6500	08-OCT-02	Manager	20	707
701	Deepak	8000	05-JAN-01	Analyst	30	707
702	Naresh	9000	10-JAN-01	Manager	10	707
703	Sumesh	7000	05-FEB-01	Salesman	20	705
704	Aditya	9000	27-NOV-07	Analyst	30	707
706	Amit		04-NOV-04	Clerk	10	702
708	Sumit	8000	06-JAN-06	Accountant	10	702

```
8 rows selected.
```

Write a query to get all employee records against all the department records (Cross join)

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
SQL> select * from dept_22co11, emp_22co11;
```

DID	DNAME	LOC	MGRID	EID	NAME	SALARY	HIRE_DATE	JOB	DID	MID
10	Accounts	Bangalore	702	707	Vishal	9500	01-JAN-01	Manager	30	
10	Accounts	Bangalore	702	705	Lalit	6500	08-OCT-02	Manager	20	707
10	Accounts	Bangalore	702	701	Deepak	8000	05-JAN-01	Analyst	30	707
10	Accounts	Bangalore	702	702	Naresh	9000	10-JAN-01	Manager	10	707
10	Accounts	Bangalore	702	703	Sumesh	7000	05-FEB-01	Salesman	20	705
10	Accounts	Bangalore	702	704	Aditya	9000	27-NOV-07	Analyst	30	707
10	Accounts	Bangalore	702	706	Amit		04-NOV-04	Clerk	10	702
10	Accounts	Bangalore	702	708	Sumit	8000	06-JAN-06	Accountant	10	702
20	Sales	Delhi	705	707	Vishal	9500	01-JAN-01	Manager	30	
20	Sales	Delhi	705	705	Lalit	6500	08-OCT-02	Manager	20	707
20	Sales	Delhi	705	701	Deepak	8000	05-JAN-01	Analyst	30	707

20	Sales	Delhi	705	702	Naresh	9000	10-JAN-01	Manager	10	707
20	Sales	Delhi	705	703	Sumesh	7000	05-FEB-01	Salesman	20	705
20	Sales	Delhi	705	704	Aditya	9000	27-NOV-07	Analyst	30	707
20	Sales	Delhi	705	706	Amit		04-NOV-04	Clerk	10	702
20	Sales	Delhi	705	708	Sumit	8000	06-JAN-06	Accountant	10	702
30	Research	Pune	707	707	Vishal	9500	01-JAN-01	Manager	30	
30	Research	Pune	707	705	Lalit	6500	08-OCT-02	Manager	20	707
30	Research	Pune	707	701	Deepak	8000	05-JAN-01	Analyst	30	707
30	Research	Pune	707	702	Naresh	9000	10-JAN-01	Manager	10	707
30	Research	Pune	707	703	Sumesh	7000	05-FEB-01	Salesman	20	705
30	Research	Pune	707	704	Aditya	9000	27-NOV-07	Analyst	30	707

30	Research	Pune	707	706	Amit		04-NOV-04	Clerk	10	702
30	Research	Pune	707	708	Sumit	8000	06-JAN-06	Accountant	10	702
40	Developing	Hyderabad		707	Vishal	9500	01-JAN-01	Manager	30	
40	Developing	Hyderabad		705	Lalit	6500	08-OCT-02	Manager	20	707
40	Developing	Hyderabad		701	Deepak	8000	05-JAN-01	Analyst	30	707
40	Developing	Hyderabad		702	Naresh	9000	10-JAN-01	Manager	10	707
40	Developing	Hyderabad		703	Sumesh	7000	05-FEB-01	Salesman	20	705
40	Developing	Hyderabad		704	Aditya	9000	27-NOV-07	Analyst	30	707
40	Developing	Hyderabad		706	Amit		04-NOV-04	Clerk	10	702
40	Developing	Hyderabad		708	Sumit	8000	06-JAN-06	Accountant	10	702

32 rows selected.

Write a query to Display EID and DName of all employees by joining over DID. (Equi join)

```
SQL> select EID, DName from emp_22co11
2 E inner join dept_22co11 D on E.DID=D.DID;
```

EID	DNAME

707	Research
705	Sales
701	Research
702	Accounts
703	Sales
704	Research
706	Accounts
708	Accounts

8 rows selected.

Display EID and DName of employees by joining over MID. (By using left outer join,

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

Right outer join and Full outer join).

```
SQL> select EID, Dname from emp_22co11 E LEFT OUTER JOIN dept_22co11 D on E.MID=D.MGRID;

  EID DNAME
-----
  708 Accounts
  706 Accounts
  703 Sales
  704 Research
  702 Research
  701 Research
  705 Research
  707

8 rows selected.

SQL> select EID, Dname from emp_22co11 E RIGHT OUTER JOIN dept_22co11 D on E.MID=D.MGRID;

  EID DNAME
-----
  705 Research
  701 Research
  702 Research
  703 Sales
  704 Research
  706 Accounts
  708 Accounts
    Developing

8 rows selected.

SQL> select EID, Dname from emp_22co11 E FULL OUTER JOIN dept_22co11 D on E.MID=D.MGRID;

  EID DNAME
-----
  707
  705 Research
  701 Research
  702 Research
  703 Sales
  704 Research
  706 Accounts
  708 Accounts
    Developing

9 rows selected.
```

5. Display the name of employees and name of their managers. (Self join)

AGNEL INSTITUTE OF TECHNOLOGY AND DESIGN

```
SQL> select E.name,M.name
  2   from emp_22co11 E,emp_22co11 M
  3  where E.EID=M.MID;
```

NAME	NAME
Vishal	Lalit
Vishal	Deepak
Vishal	Naresh
Lalit	Sumesh
Vishal	Aditya
Naresh	Amit
Naresh	Sumit

7 rows selected.

6. Display EID and DName of all employees by joining over DID (Natural Join)

```
SQL> select EID,Dname from dept_22co11 natural join emp_22co11;
```

EID	DNAME
707	Research
705	Sales
701	Research
702	Accounts
703	Sales
704	Research
706	Accounts
708	Accounts

8 rows selected.